Advanced Digital Image Processing
Lecture #11

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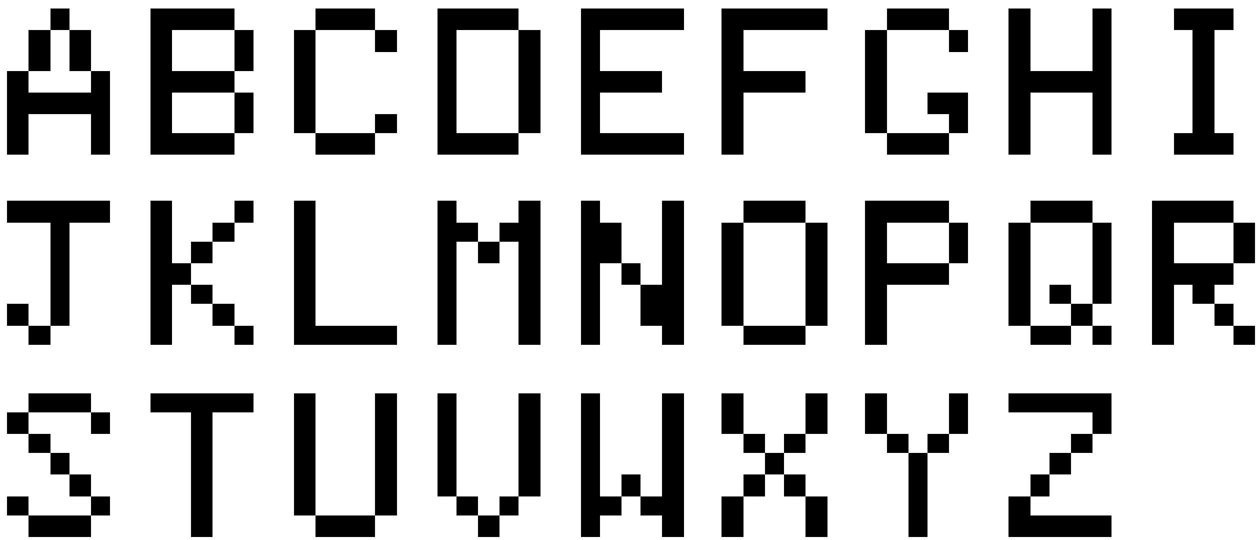
Introduction to OCR

- A machine that reads banking checks can process many more checks than a human being in the same time.
- This kind of application saves time and money, and eliminates the requirement that a human perform such a repetitive task.
- A device is to be designed and trained to recognize the 26 letters of the alphabet.

Introduction to OCR (Representation of 26 Letters)

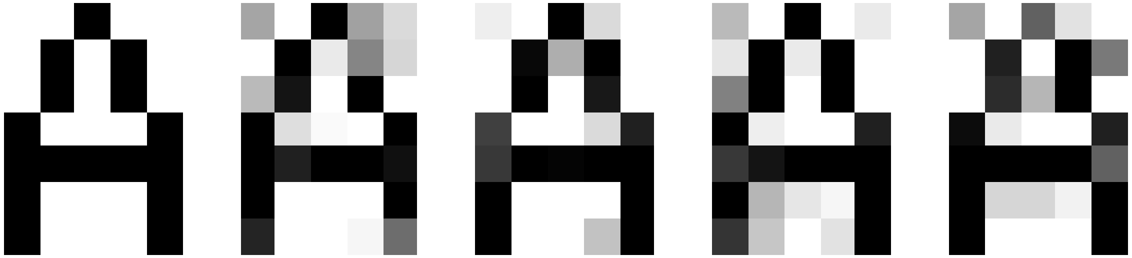
- We assume that some imaging system digitizes each letter entered in the system's field of vision. The result is that each letter is represented as a 5 by 7 grid of real values.



The ``perfect'' pictures of all 26 letters with a resolution of 5 x 7.

Introduction to OCR (Example of Noisy Versions)

- However, the imaging system is not perfect and the letters may suffer from noise:



one ``perfect'' picture of the letter ``A'' and four noisy versions (standard deviation of 0.2).

- Perfect classification of ideal input vectors is required, and
- more important reasonably accurate classification of noisy vectors are also required

Neural Network for OCR

- The twenty-six 35-element input vectors are defined as a matrix of input vectors (called alphabet).
- The target vectors are also defined in this file with a variable called targets.
- Each target vector is a 26-element vector with a 1 in the position of the letter it represents, and 0's everywhere else.
- For example, the letter ``C'' is to be represented by a 1 in the third element (as ``C'' is the third letter of the alphabet), and 0's everywhere else.

Using a Neural Network to Solve The Problem

- The network receives the 5 x 7 real values as a 35-element input vector.
- It is then required to identify the letter by responding with a 26-element output vector. The 26 elements of the output vector each represent a letter.
- To operate correctly, the network should respond with a 1 in the position of the letter being presented to the network. All other values in the output vector should be 0.
- In addition, the network should be able to handle noise.

Network Architecture (1 of 2)

- The neural network needs 35 inputs and 26 neurons in its output layer to identify the letters.
- The network is a two-layer network.
- The **log-sigmoid** transfer function at the output layer was picked because its output range (0 to 1) is perfect for learning to output Boolean values.
- The hidden layer has 10 neurons. This number was picked by **guesswork** and experience. If the network has trouble learning, then neurons can be added to this layer.

Network Architecture (2 of 2)

- The network is trained to output a 1 in the correct position of the output vector and to fill the rest of the output vector with 0's.
- However, noisy input vectors may result in the network not creating perfect 1's and 0's.
- After the network is trained the output is passed through the competitive transfer function (compet).
- This makes sure that the output corresponding to the letter most like the noisy input vector takes on a value of 1, and all others have a value of 0.
- The result of this post-processing is the output that is actually used.

Network Training (1 of 2)

- To create a network that can handle noisy input vectors it is best to train the network on both ideal and noisy vectors. To do this, the network is first trained on ideal vectors until it has a low sum-squared error.
- Then, the network is trained on 10 sets of ideal and noisy vectors.
- The network is trained on two copies of the noise-free alphabet at the same time as it is trained on noisy vectors. The two copies of the noise-free alphabet are used to maintain the network's ability to classify ideal input vectors.
- Unfortunately, after the training described above the network may have learned to classify some difficult noisy vectors at the expense of properly classifying a noise-free vector. Therefore, the network is again trained on just ideal vectors. This ensures that the network responds perfectly when presented with an ideal letter.
- All training is done using back propagation with both adaptive learning rate and momentum with the function (traingdx).

Network Training (without noise)

- The network is initially trained without noise for a maximum of 5000 epochs or until the network sum-squared error falls beneath 0.1.

```
P = alphabet;   T = targets;
net.performFcn = 'sse';
net.trainParam.goal = 0.1;
net.trainParam.show = 20;
net.trainParam.epochs = 5000;
net.trainParam.mc = 0.95;
[net,tr] = train(net,P,T);
```

Network Training (with noise)

- To obtain a network not sensitive to noise, we trained with two ideal copies and two noisy copies of the vectors in alphabet.
- The target vectors consist of four copies of the vectors in target. The noisy vectors have noise of std 0.1 and 0.2 added to them.
- This forces the neuron to learn how to properly identify noisy letters, while requiring that it can still respond well to ideal vectors.
- To train with noise, the maximum number of epochs is reduced to 300 and the error goal is increased to 0.6, reflecting that higher error is expected because more vectors (including some with noise), are being presented.

Network Training (with noise)

- netn = net;
- netn.trainParam.goal = 0.6;
- netn.trainParam.epochs = 300;
- T = [targets targets targets targets];
- for pass = 1:10
- P = [alphabet, alphabet, …
  (alphabet + randn(R,Q)*0.1), ...
  (alphabet + randn(R,Q)*0.2)];
- [netn,tr] = train(netn,P,T); end

Network Training (without noise)

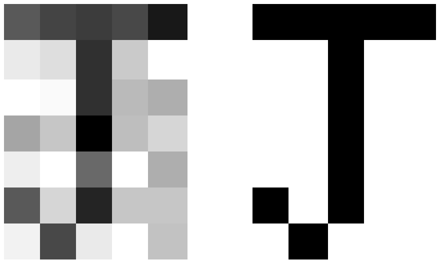
- Once the network is trained with noise, it makes sense to train it without noise once more to ensure that ideal input vectors are always classified correctly.
- Therefore, the network is again trained with code identical to the Training Without Noise section.

Estimating the System Performance

- The reliability of the neural network pattern recognition system is measured by testing the network with hundreds of input vectors with varying quantities of noise.
- For example we create a noisy version (SD 0.2) of the letter ``J" and query the network:
- noisyJ = alphabet(:,10)+randn(35,1) * 0.2; plotchar(noisyJ); A2 = sim(net,noisyJ); A2 = compet(A2); answer = find(compet(A2) == 1); plotchar(alphabet(:,answer));

Example of Noise

- Here is the noisy letter and the letter the network picked (correctly).

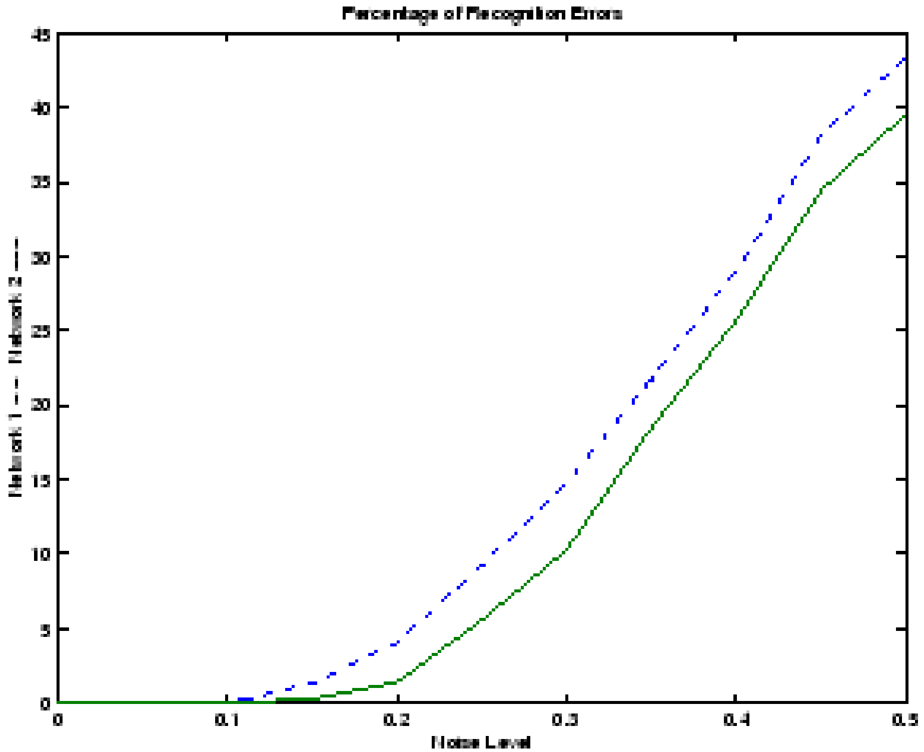


The network is tested with a noisy version of the letter ``J''.

Performance

- The script file appcr1 tests the network at various noise levels, and then graphs the percentage of network errors versus noise.
- Noise with a mean of 0 and a standard deviation from 0 to 0.5 is added to input vectors.
- At each noise level, 100 presentations of different noisy versions of each letter are made and the network's output is calculated.
- The output is then passed through the competitive transfer function so that only one of the 26 outputs (representing the letters of the alphabet), has a value of 1.
- The number of erroneous classifications is then added and percentages are obtained.

Performance for The Network (Trained with and Without Noise)



Performance

- The solid line on the graph shows the reliability for the network trained with and without noise.
- The reliability of the same network when it had only been trained without noise is shown with a dashed line. Thus, training the network on noisy input vectors greatly reduces its errors when it has to classify noisy vectors.
- The network did not make any errors for vectors with noise of std 0.00 or 0.05. When noise of std 0.2 was added to the vectors both networks began making errors.
- If a higher accuracy is needed, the network can be trained for a longer time or retrained with more neurons in its hidden layer. Also, the resolution of the input vectors can be increased to a 10-by-14 grid. Finally, the network could be trained on input vectors with greater amounts of noise if greater reliability were needed for higher levels of noise.

Student Assignment (G3)

- What is Shading?
- Types of Shading:
  - Constant Shading
  - Interpolated Shading
  - Accurate Shading
  - Flat Shading
  - Phong Shading
- Distinguish between illumination and shading models
- Shading problems

List of Reference Books

- Rafael C. Gonzalez, “Digital Image Processing”.
- Anil K. Jain, “Fundamental of Digital Image Processing”.
- Dana H. Ballard, “Computer Vision”
- David H. Hubel, “Eye, Brain, and Vision”
- Dwayne Phillips, “Image Processing in C”